

Question:
The feedback loop in men which regulates the production of sex hormones includes all but which of the following?

- Answer Choices:
- A. follicle-stimulating hormone
 - B. oxytocin
 - C. pituitary gland
 - D. hypothalamus
 - E. gonadotropin-releasing hormone
 - F. luteinizing hormone
 - G. testosterone

Reasoning + Answer:

<think>

The hypothalamic-pituitary-gonadal (HPG) axis plays a crucial role in regulating sex hormone production in men. The negative feedback loop within this system ensures that hormone levels remain within a narrow range, preventing both deficiency and excess. The hypothalamus releases gonadotropin-releasing hormone (GnRH), which stimulates the pituitary gland to release follicle-stimulating hormone (FSH) and luteinizing hormone (LH). These hormones then stimulate the gonads to produce testosterone and other sex hormones. The negative feedback loop involves the production and reduction of the hormones. For instance, when testosterone levels increase, it provides a negative feedback signal to the hypothalamus, reducing GnRH secretion and thus reducing FSH and LH secretion, which in turn reduces testosterone production. Oxytocin is not directly involved in the feedback loop regulating sex hormone production. It is primarily involved in social bonding and childbirth, but is not directly related to the regulation of testosterone levels in the testes. The pituitary gland and gonadotropin-releasing hormone are directly involved in the negative feedback regulation of sex hormone production.

</think>

<answer>A.